3 *Logical structure of meaning*

Linguistic theory distinguishes between *semantic meaning* of a sentence or an expression and additional *pragmatic enrichments* to the literal meaning. The literal meaning is the context-independent logical core meaning. Pragmatic enrichments arise in context by taking into account general world knowledge or the reasons why a speaker used language in the way that they did. Pragmatic enrichments may be part of what we "perceive" to be the most likely interpretation of a sentence; they may be what the speaker meant to say, but not what the sentence means literally.[10]

Our current goal is to learn to excavate the logical structure of natural language sentences. Therefore, we have to learn to "strip off the pragmatic layer," so to speak. Indeed, the logical operators of PropLog bear a close correspondence to natural language expressions *and*, *or*, *if* and *if and only if*. But the correspondence is not a perfect match. This is partly due to the difference between semantic meaning and pragmatic enrichment.[11] Let us look at a few interesting cases.

[10] The most obvious example is an utterance of a sentence like *The weather is so nice today!* when clearly meant ironically.

[11] There are also arguments suggesting that the logical operators of PropLog are *not* good enough for analyzing the semantic meaning of their corresponding natural language expressions. Most importantly, this concerns the semantic meaning of conditional sentences, i.e., natural language sentences with *if . . . , then . . . .*

3.1 *Conjunction & order-sensitivity*

Logical conjunction is order-insensitive: $\varphi \wedge \psi$ and $\psi \wedge \varphi$ are logically equivalent. But natural language conjunctions are not necessarily. Saying that

They got married and had children.

might be understood differently from saying that

They had children and got married.

Still, we would analyze both of these sentences as having a logical-conjunctive meaning, additional inferences about the temporal order of events notwithstanding.

3.2 *Disjunction: inclusive vs. exclusive*

Logical disjunction is *inclusive*: $\varphi \vee \psi$ is true when φ and ψ are both true. Yet natural language disjunctions *can* be understood as exclusive disjunction according to which exactly one, but not both disjuncts are true. For example, if I say:

She owns a Porsche or a Ferrari.

you might take that to mean that she doesn't own both. But did I *literally* say that? — The logical meaning of that sentence is more plausibly just analyzed as inclusive disjunction. This is because it is not contradictory to say:

She owns a Porsche or a Ferrari and possibly both.

This contrasts with the weirdness of saying:

She owns either *a Porsche or a Ferrari and possibly both.*

suggesting that natural language *either . . . or . . . does* have an exclusive disjunctive meaning, but simple *or* does not.

3.3 Conditional perfection

PropLog analyzes implication as so-called *material implication*, but it is controversial whether natural language conditional sentences might not have a much richer meaning, e.g., requiring a discernible causal-inferential relation between antecedent and consequent. Moreover, similar to the case of inclusive vs. exclusive disjunctions, many natural language conditionals receive a so-called *conditional perfection reading*, i.e., they are read like an "if and only if" statement which would correspond to logical operator $\leftrightarrow$, even though their logical meaning might just be that of a simple implication $\rightarrow$. For example, the conditional statement:

If Bubu comes, Kiki comes.

does suggest to a certain degree that Kiki *only* comes when Bubu does. But that is not the logical core meaning of this sentence. It is rather a pragmatic enrichment on top of the logical meaning of the sentence. To see this, compare the difference in weirdness between saying

If Bubu comes, Kiki comes, and maybe Kiki comes no matter what.

which seems fine, especially when compared to the much weirder:

Kiki comes only if Bubu comes, and maybe Kiki comes not matter what.

3.4 Excavating the logical structure of natural language sentences

Despite superficial discrepancies, it is possible and highly informative to try to lay bare the logical structure of natural language sentences. The logical relation between propositions contained in a natural language sentence may sometimes be opaque. Take the following example sentence:

If drawing a red card means that I lost, I lost.

This sentence does have a propositional-logical structure, even though it is not at all apparent from the way it is realized in natural language. There are two atomic (truth-evaluable) propositions involved here:

p: I drew a red card. q: I lost.

With this *translation key*, we can then determine the logical structure of the sentence above as:

$$(p \rightarrow q) \rightarrow q$$

Here are a few other examples.

(i) John is hungry.
Key: p: John is hungry. *Logical form:* p

(ii) Mary likes John.
Key: p: Mary likes John. *Logical form:* p

(iii) John is hungry and Mary likes John.
Key: p: John is hungry. q: Mary likes John.
Logical form: $p \wedge q$

(iv) Berlin is east of Hamburg and Bremen.
Key: p: Berlin is east of Hamburg. q: Berlin is east of Bremen.
Logical form: $p \wedge q$

(v) Call me 'sweetie' once more and, no kidding, I'll step on your toe.
Key: p: I'm not kidding. q: I'll step on your toe.
r: You call me 'sweetie' once more.
Logical form: $p \wedge (r \rightarrow q)$

(vi) I will only play salsa, if you give me earplugs.[12]
Key: p: I will play salsa. q: You give me earplugs.
Logical form: $p \rightarrow q$

Exercise 5. For each of the following sentences give the translation key and the (propositional) logical form.

(i) If John is hungry, then Mary likes John.

(ii) If John is hungry and Bill runs, then Mary doesn't likes John.

(iii) John is hungry or it's false that Bill runs.

(iv) Mary likes John or Mary doesn't like John.

(v) Mary likes John but John is hungry.

(vi) Mary likes John provided Bill runs.

[12] There is a likely inclination to analyze this as $p \leftrightarrow q$, but it is not clear whether this is part of the semantic/logical content here. Suppose you know that the sentence is true. Now you learn that earplugs have been handed over. Does that *logically* entail that salsa was played? No, because the sentence only says that earplugs are a *necessary* but not necessarily *sufficient* condition for salsa.

4 Valuation functions, possible worlds & logical entailment

4.1 Valuation functions

The truth-tables method introduced above is very practical, but a proper definition of the semantics of PropLog should better not rely on vocabulary like "rows" and "columns," but use mathematical notions like "function" or "set." The proper definition of the semantics of PropLog is therefore formulated in terms of so-called valuation functions.

A *valuation function V* is a function that assigns to every formula of propositional logic a unique truth value (0 or 1). More specifically, think of a valuation function as a way of assigning truth-values to all proposition letters. Once we know the truth or falsity of all proposition letters, the truth or falsity of all complex formulas is non-arbitrary, because it is fixed by the meaning of the sentential connectives.

$V(p), V(q), \ldots$		arbitrary
$V(\neg\phi) = 1$	iff	$V(\phi) = 0$
$V(\phi \wedge \psi) = 1$	iff	$V(\phi) = 1$ and $V(\psi) = 1$
$V(\phi \vee \psi) = 1$	iff	$V(\phi) = 1$ or $V(\psi) = 1$
$V(\phi \rightarrow \psi) = 0$	iff	$V(\phi) = 1$ and $V(\psi) = 0$
$V(\phi \leftrightarrow \psi) = 1$	iff	$V(\phi) = V(\psi)$

Table 6: Semantics of propositional logic in terms of valuation functions

A valuation function is analogous to a row in a truth table.

4.2 Possible worlds & truth-conditional meaning

A valuation function can also be thought of as a *possible world*.[13] A possible world is a way in which things could be. A possible world conforms to the rules of logic, but otherwise allows all atomic propositions to be different from the facts in the *actual world*.

[13] This terminology is widely used in formal semantics and philosophy.

Using valuation functions or, equivalently, possible worlds, we can give a clear definition of the meaning of a formula (and, therefore by extension, to the logical meaning of a natural language sentence). The truth-conditional meaning of formula φ is the set of all valuation functions, or possible worlds, for which φ is true. Formal semantics often uses notation $[\![\varphi]\!]$ to denote the meaning of a formula in this sense:

$$[\![\varphi]\!] = \{V \mid V \text{ is a valuation function such that } V(\varphi) = 1\}$$

Alternatively, we could say that the meaning of a formula is the set of possible worlds in which it is true.[14]

[14] Compare the famous passage from Wittgenstein's *Tractatus Logica Philosophicus*: "Einen Satz verstehen, heißt, wissen was der Fall ist, wenn er wahr ist. (Man kann ihn also verstehen, ohne zu wissen, ob er wahr ist.)" [TLP 4.024]

4.3 Tautologies, contradictions, contingencies & equivalence (revisited)

Using valuation functions and/or possible worlds, we can also give a proper definition of notions we used previously in the context of truth tables. We

say that φ is a tautology iff $V(\varphi) = 1$ for all valuation functions V. We say that φ is a contradiction iff $V(\varphi) = 0$ for all valuation functions V. We say that φ is a contingency iff there is at least one valuation function V_1 such that $V_1(\varphi) = 1$ and there is at least one valuation function V_2 such that $V_2(\varphi) = 0$. We say that φ and ψ are logically equivalent, iff for all valuation functions $V(\varphi) = V(\psi)$.

4.4 Logical entailment, informativity, & independence

We can now also give a definition of logical entailment. Formula φ logically entails ψ iff for all valuation functions V, if $V(\varphi) = 1$, then $V(\psi) = 1$. An alternative formulation is this: φ logically entails ψ iff $[\![\varphi]\!] \subseteq [\![\psi]\!]$.

Based on the notion of logical meaning in terms of sets of possible worlds, we can also tackle another otherwise extremely elusive concept, namely that of *informativity*, i.e., the amount of information given. We say that a contingency φ is at least as informative as ψ iff $[\![\varphi]\!] \subseteq [\![\psi]\!]$, and that φ is strictly more informative than ψ iff $[\![\varphi]\!] \subset [\![\psi]\!]$.

This notion of logical informativity is a relative one. It does not give an absolute measure of information for each formula, but only compares two formulas in terms of their relative informativity. It also only compares some formulas; in fact, the ordering induced by this definition of logical informativity is a *partial order*.[15]

[15] Probabilistic notions of information, e.g., as given by information theory extend the notion of logical information to an absolute measure of information which yields a total order.

Formulas φ and ψ are *logically independent* iff all of the following are non-empty: $[\![\varphi]\!] \cap [\![\psi]\!]$, $[\![\neg\varphi]\!] \cap [\![\psi]\!]$, $[\![\varphi]\!] \cap [\![\neg\psi]\!]$, $[\![\neg\varphi]\!] \cap [\![\neg\psi]\!]$. Intuitively, logical independence means that for every possible truth-value assignment to both φ and ψ is possible: there exists at least one valuation function (possible world) for each combination of truth values for φ and ψ. In other words, the truth value of φ does not determine the truth value of ψ and vice versa. Just learning the truth value of one formula does not reveal the truth value of the other.

4.5 Argument schemas & validity

The important concept of logical validity can be thought of, at least for PropLog, as an extension of the notion of logical entailment to more than two formulas. Concretely, we define logical validity of so-called argument schemas. If $\varphi_1, \varphi_2, \ldots, \varphi_n$ and ψ are forumlas of propositional logic, then $\varphi_1, \varphi_2, \ldots, \varphi_n / \psi$ is an ARGUMENT SCHEMA. Formulas $\varphi_1, \varphi_2, \ldots, \varphi_n$ are the PREMISSES, formula ψ is the CONCLUSION.

The argument schema $\varphi_1, \varphi_2, \ldots, \varphi_n / \psi$ is VALID iff for all valuation functions V such that $V(\varphi_1) = V(\varphi_2) = \cdots = V(\varphi_n) = 1$ it also holds that $V(\psi) = 1$. If valid, we write $\varphi_1, \varphi_2, \ldots, \varphi_n \models \psi$.

We can use truth tables again to check whether any given argument schema is valid or not. Consider the argument schema: $\varphi \vee \psi, \neg\psi \,/\, \varphi$. A natural language analogue would be the following. We know or assume the

following premisses:

> The murderer is the butler or the gardener.
> The gardener is not the murderer.

And the putative conclusion is:

> The butler is the murderer.

We would like to find out whether the conclusion logically follows from the premisses. Table 7 tells us that it does: in all rows (=valuation functions or possible worlds) in which all the premisses are true, which are marked with an asterisk (*) in Table 7, the conclusion (in the last column) is also true.

φ	ψ	$\varphi \vee \psi$	$\neg\psi$	/	φ
1	1	1	0		1
1	0	1	1	*	1
0	1	1	0		0
0	0	0	1		0

Table 7: Truth table for checking the validity of argument schema $\varphi \vee \psi, \neg\psi \,/\, \varphi$

Table 8 shows a more extensive example of the truth-table method for the argument schema $\neg\varphi \rightarrow (\psi \wedge \neg\chi), \neg\psi \rightarrow \neg\chi \,/\, \varphi$, which is not valid. The table also introduces names for each row, in terms of (types of) possible worlds, so that we can conveniently say that a possible world like w_6 in which φ and χ are false and ψ is true is a counterexample to validity: both premises are true, but the conclusion is false.

	φ	ψ	χ	$\neg\varphi$	$\neg\chi$	$\psi \wedge \neg\chi$	$\neg\varphi \rightarrow (\psi \wedge \neg\chi)$	$\neg\psi$	$\neg\psi \rightarrow \neg\chi$	/	φ	
w_1	1	1	1	0	0	0	1	0	1	*	1	
w_2	1	1	0	0	1	1	1	0	1	*	1	
w_3	1	0	1	0	0	0	1	1	0		1	
w_4	1	0	0	0	1	0	1	1	1	*	1	
w_5	0	1	1	1	0	0	0	0	1		0	
w_6	0	1	0	1	1	1	1	0	1	*	0	←
w_7	0	0	1	1	0	0	0	1	0		0	
w_8	0	0	0	1	1	0	0	1	1		0	

Table 8: Example of the truth-table method applied to an argument schema that is not valid

4.6 *Theorems of propositional logic*

Based on the definitions we have seen so far, we can prove illuminating facts about PropLog.[16]. Here are some examples.

Proposition 1. If φ is a tautology, then $\neg\varphi$ is a contradiction.

Proof. If φ is a tautology, then for all valuation functions V we have $V(\varphi) = 1$. For any valuation function V if $V(\varphi) = 1$ then, by definition, $V(\neg\varphi) = 0$. Therefore, $V(\neg\varphi) = 0$ for all V, which means that φ is a contradiction. □

[16] Proving properties of a logical system is a special subfield called *metalogic*

Proposition 2. $\varphi \to \psi$ is a tautology iff φ logically entails ψ.[17]

Proof. First assume that $\varphi \to \psi$ is a tautology. We need to show that it follows that φ logically entails ψ. Towards contradiction, let us assume that φ does not logically entail ψ. This can only be the case if there is at least one valuation function V for which $V(\varphi) = 1$ and $V(\psi) = 0$. For this valuation function, by definition of the meaning of $\to$, it holds that $V(\varphi \to \psi) = 0$. So, we contradict our assumption that $\varphi \to \psi$ is a tautology.

Next, we assume that φ logically entails ψ. We need to show that $\varphi \to \psi$ is a tautology. If φ logically entails ψ, this means that there is no valuation function V for which $V(\varphi) = 1$ and $V(\psi) = 0$. But, from the definition of the semantics of $\to$, we know that the only possibility for $\varphi \to \psi$ to be false is if $V(\varphi) = 1$ and $V(\psi) = 0$. Since this is ruled out by our assumption, $\varphi \to \psi$ must be a tautology. □

[17] Notice that this may seem trivial, but there are really two notions that are related by this proposition. One is the definition of the sentential connective $\to$, the other is a high-level notion of logical entailment defined on top of the semantics for *all* formulas.

We can even generalize this last result to show that propositional logic is very well-behaved in the sense that entailment, truth of conditional statements and validity all coalesce. This need not be the case for any logical system, and so you might want to try to prove the following for yourself:

Proposition 3. $(\varphi_1 \wedge \cdots \wedge \varphi_n) \to \psi$ is a tautology iff $\varphi_1 \wedge \cdots \wedge \varphi_n$ entails ψ iff $\varphi_1, \ldots, \varphi_n \models \psi$.

Exercise 6. Use the truth-table method to check whether the following argument schemas are valid:

(i) $p \to q, p \,/\, q$ [modus ponens]

(ii) $p \to q, q \,/\, p$ [affirmation of the consequent]

(iii) $p \to q, \neg q / \neg p$ [modus tollens]

(iv) $\neg(p \wedge q), \neg q \,/\, p$

Exercise 7. For each of the following claims, determine whether it is true or false. If it is true prove it. If it is false, give a refutation by counterexample.

(i) If φ is a tautology and ψ is a contradiction, $\varphi \to \psi$ is a contradiction.

(ii) If ψ is a tautology, $\varphi \to \psi$ is a tautology.

(iii) If φ is a tautology, $\varphi \to \psi$ is a tautology.

(iv) If $\varphi \vee \psi$ is a contradiction, φ is a contradiction.